Healthcare *Management*

CUSTOMER REQUIREMENTS AND SCOPE

Health-Center *Inventory* Management

Medicine stock, requisitions, expired returns and patient registration for health centers, on Sky ERP.

First customer	Mr. Emmanuel (pilot)
Prepared by	SkyEngPro
Version	0.3, draft for validation with the customer

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1. Purpose

Health centers manage their medicine stock on paper today. This project gives them a simple application to record every medicine that comes in, is given to a patient, or is returned because it expired, so that the stock is always known and every movement can be traced to a person.

The solution reuses existing Sky ERP modules and adds a simplified screen set made for health center staff, not for ERP specialists.

What success looks like

- Staff always know how much of each medicine is in stock.
- Every stock change shows who did it, when, how much and why.
- Medicines are ordered on time, before they run out.
- Expired medicine is never given to a patient and is returned in an orderly way.
- The pilot health center works for one full month without the paper register.

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2. Current situation

AREA	TODAY
Records	Manual registers kept by each health center
Ordering	Requests are made directly to Mr. Emmanuel
Where medicines come from	The store, the open market, medical delegates, the delegation
Giving medicine to patients	Written in the stock register by hand
Expired medicine	Returned to Mr. Emmanuel for disposal

Problems to solve

1. Stock is kept in manual records, which are hard to search and easy to get wrong.
2. Stock movements are hard to track, so losses and shortages are found late.
3. Orders are not always placed on time.

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3. Scope

Included (confirmed)

- **Drug requisition:** a health center asks for medicines.
- **Dispense drug:** giving medicine to a patient, which reduces the stock. Every dispensing is linked to a registered patient.
- **Return expired:** sending expired medicine back, which reduces the stock.
- **Patient registration:** registering and finding patients.

Included to make the above work (proposed)

- A list of medicines with a minimum stock level.
- **Receiving stock**, with where it came from, its batch and expiry date.
- Alerts for low stock and for medicine about to expire.
- Users with simple roles, and reports.
- Entering the opening stock from the first physical count.

Not included

- Billing, prices charged to patients, payments and insurance.
- Purchasing from suppliers and accounting.
- Anything other than medicine stock and patient registration.

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4. Users and roles

ROLE	WHO	WHAT THEY CAN DO
Staff	Nurses, pharmacists, store keepers	Register patients, receive stock, dispense medicines, ask for medicines, record expired returns, see reports
Administrator	Health center manager, Mr. Emmanuel TBC	Everything staff can do, plus approve or reject requests, manage users, and cancel a wrong entry

TBC: whether staff may also cancel a wrong entry or adjust the stock, or administrators only.

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5. How it will work

Ordering medicines

1. The system shows medicines that are running low.
2. Staff ask for the medicine they need, with quantity and urgency.
3. An administrator approves or rejects the request.
4. The approved request reaches Mr. Emmanuel TBC: inside the system, or as a printed or PDF document.
5. When the medicines arrive they are received into stock.

Receiving stock. Staff enter the medicine, quantity, batch number, expiry date and where it came from: store, open market, medical delegate, delegation or other.

Dispensing to a patient. Staff open the patient's file and dispense the medicines. The system uses the batch that expires first, refuses expired medicine, and refuses a quantity larger than what is available. A wrong entry can be cancelled with a reason, and the stock goes back.

Returning expired medicine. The system lists expired medicine and medicine expiring soon. Staff choose the batch and quantity to return and who receives it TBC. The stock goes down and the return is recorded.

Registering a patient. Staff search first to avoid duplicates, then register the patient. The patient's file shows the medicines given to them.

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6. Requirements

6.1 Medicines

REQUIREMENT	PRIORITY	BASIS
Add and edit a medicine: name, category, form, strength, unit	Must	Proposed
Set a minimum stock level; show "to restock" when it is reached	Must	Proposed
Search and filter medicines by name, category, status and expiry	Must	Proposed
Show for each medicine its stock, nearest expiry, batches and full history	Must	Proposed
Archive a medicine without losing its history	Should	Proposed
Import the first medicine list from a spreadsheet	Should	Proposed

6.2 Receiving stock

REQUIREMENT	PRIORITY	BASIS
Receive a medicine with quantity, batch number, expiry date and date	Must	Proposed
Record where each receipt came from: store, open market, medical delegate, delegation, other	Must	Confirmed
Enter the opening stock from the first physical count	Must	Proposed
Record the delivery or receipt number TBC	Should	Proposed
Receive several batches of one medicine in a single entry	Should	Proposed

6.3 Drug requisition

REQUIREMENT	PRIORITY	BASIS
Ask for a medicine with quantity, urgency and reason	Must	Confirmed
Administrator approves or rejects, with a note	Must	Confirmed
Make an approved request available to Mr. Emmanuel TBC	Must	Confirmed
Suggest a request automatically when stock is low	Should	Proposed
Notify the administrator of waiting requests	Should	Proposed
Follow the request status: waiting, approved, rejected, received	Should	Proposed

6.4 Dispensing

REQUIREMENT	PRIORITY	BASIS
Dispense medicine to a registered patient and reduce the stock	Must	Confirmed
No medicine can be given without choosing a patient	Must	Confirmed
Never dispense expired medicine	Must	Confirmed
Use the batch that expires first	Must	Proposed
Refuse a quantity above the available stock, with a clear message	Must	Proposed
Record the user, date and time of every dispensing	Must	Confirmed
Never record the same entry twice, even after a poor connection	Must	Proposed
Cancel a wrong dispensing with a reason, returning the stock; the original entry stays visible as cancelled	Should	Proposed
Print a dispensing slip TBC	Could	Proposed

6.5 Expired medicine

REQUIREMENT	PRIORITY	BASIS
List expired medicine and medicine expiring soon (default 90 days)	Must	Proposed
Record the return of an expired batch: quantity, date and receiver	Must	Confirmed
Alert staff about medicine about to expire	Should	Proposed
Print a return slip for the receiver's signature TBC	Should	Proposed
Record damaged or lost stock with a reason	Should	Proposed

6.6 Patients

REQUIREMENT	PRIORITY	BASIS
Register a patient: first name, last name, sex, date of birth, phone	Must	Confirmed
Search by name or phone, and warn about a possible duplicate	Must	Proposed
Show the medicines given to each patient in the patient file	Must	Proposed
Edit a patient; a patient with history cannot be deleted	Must	Proposed
Optional details such as blood group and allergies TBC	Should	Proposed

6.7 Users and reports

REQUIREMENT	PRIORITY	BASIS
The administrator creates, disables and resets the password of users	Must	Proposed
Two roles, Staff and Administrator, with the rights listed in section 4	Must	Proposed
Keep a trail of who did what and when	Must	Confirmed
Dashboard: medicines, to restock, expiring soon, patients, waiting requests	Must	Proposed
Stock movements report with filters, exportable to Excel	Must	Confirmed
Current stock report and expiry report	Must	Proposed
Requests report and consumption per medicine, to help ordering	Should	Proposed
Print any report	Should	Proposed

7. Other expectations

- **Simple to use:** a new staff member can work with it after a short training.
- **Any device:** works on phones, tablets and computers.
- **Languages:** English and French.
- **Poor connection:** the application opens without internet. Receiving, dispensing and requests made offline are saved and sent once the connection returns, without duplicates TBC.
- **Security and privacy:** each person has their own login; patient information is only seen by authorized users TBC: legal requirements.
- **Nothing is erased:** stock movements are never deleted. Corrections are shown as cancelled entries.
- **Backups:** the data is backed up automatically.

8. Starting to use it

1. Collect the medicine list from the paper register.
2. Do a physical stock count, with batch and expiry dates.
3. Enter the count as opening stock.
4. Create the users and train the staff.
5. Run the application next to the paper register for an agreed period TBC.
6. Stop the paper register, and review the reports weekly during the first month.

9. Assumptions and risks

Assumptions

- The health center has a phone, tablet or computer, and occasional internet.
- The customer provides the medicine list and does the opening stock count.

Main risks

RISK	HOW WE REDUCE IT
Staff keep using the paper register	Training, a clear end date for the parallel run, weekly review
Poor internet	The application works offline for stock tasks
Opening stock count is inaccurate	Count with two people, check in the first month
Dispensing feels slower than paper	Few screens, quick patient search, test with real staff in the pilot
The scope grows (billing, prices)	Changes go through an agreed change request

10. Open questions

#	QUESTION
1	Does Mr. Emmanuel approve requests and receive expired medicine inside the system, or only on paper or PDF?
2	Is the delivery or receipt number needed for each receipt, in addition to where it came from?
3	What does an expired return need: receiver name, signature, a return slip, a disposal certificate?
4	Who may cancel a wrong entry and who may adjust the stock: staff or administrators only?
5	Which patient details are required, and are there legal rules about them?
6	Are there medicines without an expiry date, or needing special handling?
7	What internet, devices and printers does the pilot health center have?

#	QUESTION
8	Which printed documents are required: request, return slip, dispensing slip?
9	Who provides the medicine list and does the opening count, and when?
10	What is the target date for the pilot, and which health center?
11	Who signs the acceptance? We assume Mr. Emmanuel

11. Next steps

STEP	WHAT	WHO
1	Answer the open questions and validate this document	Mr. Emmanuel, SkyEngPro
2	Confirm which Sky ERP modules are reused and what must be added	SkyEngPro
3	Build and test the missing parts	SkyEngPro
4	Prepare the pilot: users, opening stock, training	SkyEngPro, health center
5	Run the pilot and review weekly	Mr. Emmanuel, SkyEngPro
6	Extend to other health centers	SkyEngPro

Dates will be set once the open questions are answered.

12. Acceptance

The pilot is accepted by Mr. Emmanuel when these checks pass in the health center:

1. Receiving 100 tablets from the delegation shows 100 in stock, with their source.
2. Dispensing 30 tablets to a registered patient leaves 70, with the patient and user recorded.
3. A dispensing larger than the stock is refused and nothing changes.
4. With two batches, the one that expires first is used.
5. Expired medicine cannot be dispensed.
6. Returning an expired batch reduces the stock and records the receiver.
7. A medicine below its minimum shows as "to restock", and a request can be raised and approved.
8. A wrong dispensing is cancelled with a reason and the stock goes back.
9. Work done offline is recorded once, not twice, when the connection returns.
10. Staff cannot approve a request.
11. The application is comfortable to use on a phone, a tablet and a computer.

This document reflects the information gathered so far and will be updated once the open questions are answered.